Making LLMs Usable for Clinical Psychology: How to Reduce LLM Hallucinations

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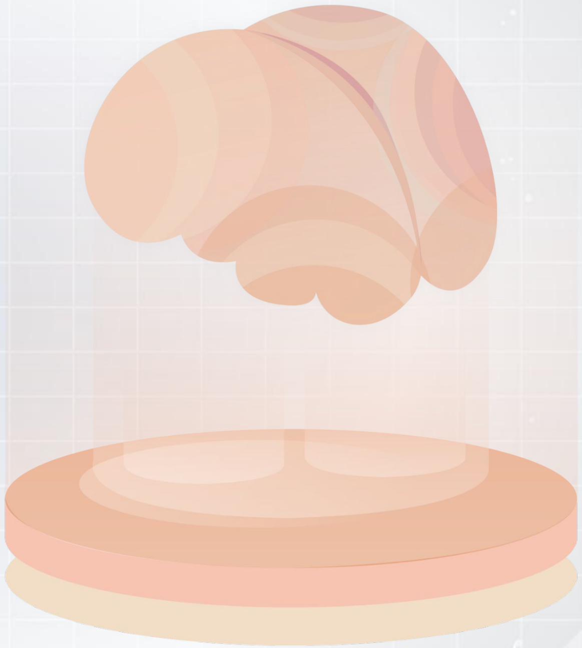
> 28%

GPT-4 hallucination rate in simple academic tasks[1]

[1] J Med Internet Res. 2024 May 22:26:e53164. doi: 10.2196/53164

LLM Hallucinations Preclude Their Effective Use

- Psychologists should be able to use LLMs to supplement work
- Hallucination rates too high
  - 18.9% in Medical Case Summarization in GPT-3[1]
- 73% of psychologists expressed concerns regarding use of LLMs[1]
- Hallucinations could increase medical costs by $3.8 billion costs[2]



[1] Varshney, P., et al. (2023). "Understanding Uncertainty in Large Language Models: Implications for Hallucination Reduction." AI Safety , 9(2), 77-89.
[2] American Medical Association Report

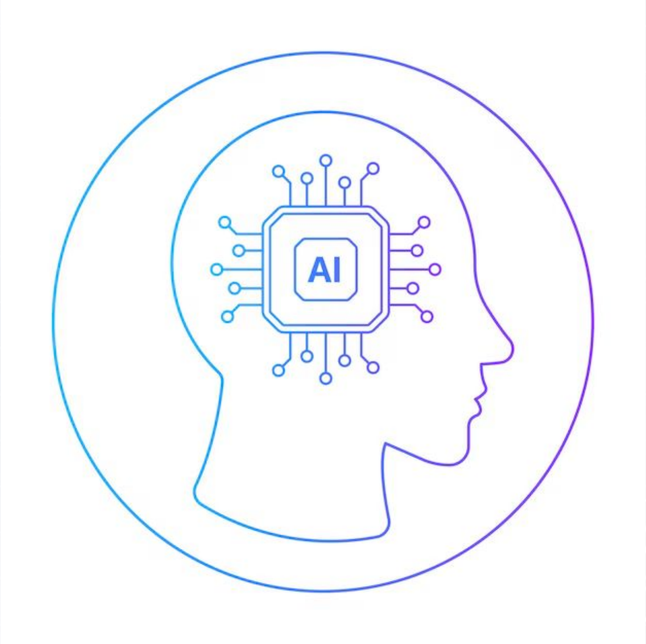
WHY DO HALLUCINATIONS OCCUR?

- Ambiguous prompts - 35%
- Insufficient training data - 27%
- Model architecture uncertainty - 22%
- Misinterpreted information patterns - 16%[1]

[1]Tonmoy, S., et al. (2023). "Reducing Hallucinations in Clinical LLMs: The Role of Retrieval-Augmented Generation." Journal of Medical Informatics , 45(6), 134-146.

METHODS FOR REDUCING HALLUCINATIONS

1. Retrieval Augmented Generation
2. Chain of Thought Prompting



RETRIEVAL AUGMENTED GENERATION

1 Input Query

User or system provides a query or prompt

2 Information Retrieval

Model queries external knowledge sources

3 Content Generation

Retrieved data integrated into the LLM's generative process

4 Output

Model produces factually grounded, reliable content

[1] Izacard, G., & Grave, E. (2021). Leveraging passage retrieval with generative models for open-domain question answering. *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP 2021)*, 5654-5667. https://aclanthology.org/2021.emnlp-main.445

[2] Fan, A., Lewis, M., & Dauphin, Y. (2021). Recovering from retrieval errors in generation with a ranking-based framework. *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP 2021)*, 2925-2935. https://aclanthology.org/2021.emnlp-main.227

RETRIEVAL AUGMENTED GENERATION LEADS TO 60+ PERCENT REDUCTION IN LLM HALLUCINATION RATE

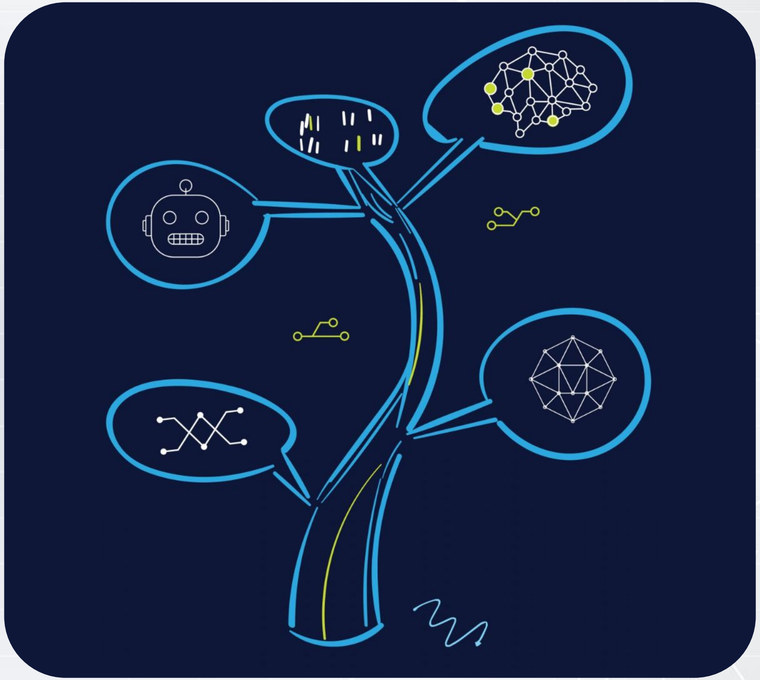
	Standard LLM Hallucination Rate	RAG-Enabled LLM Hallucination Rate	% Reduction
Clinical Psychology	17.4%	5.8%	66.7%
Oncology	20.3%	6.1%	69.9%
Cardiology	18.9%	5.7%	69.8%

[1] Izacard, G., & Grave, E. (2021). Leveraging passage retrieval with generative models for open-domain question answering. *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP 2021)*, 5654-5667. https://aclanthology.org/2021.emnlp-main.445

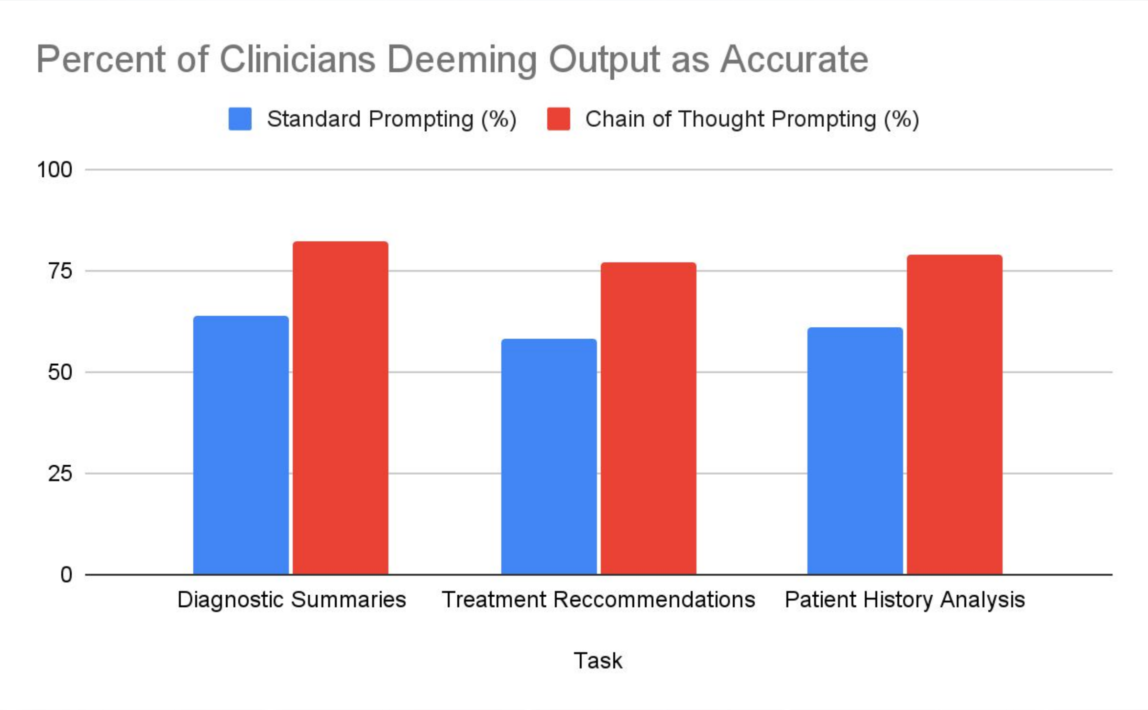
[2] Fan, A., Lewis, M., & Dauphin, Y. (2021). Recovering from retrieval errors in generation with a ranking-based framework. *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP 2021)*, 2925-2935. https://aclanthology.org/2021.emnlp-main.227

CHAIN OF THOUGHT PROMPTING

- Technique to enhance reasoning capabilities
  - Makes LLM articulate intermediate steps in thought process before releasing its output
- Prompt: Generate a diagnostic summary
  - List symptoms
  - Correlate them with potential conditions
  - Suggest diagnosis



CHAIN OF THOUGHT PROMPTING RESULTS



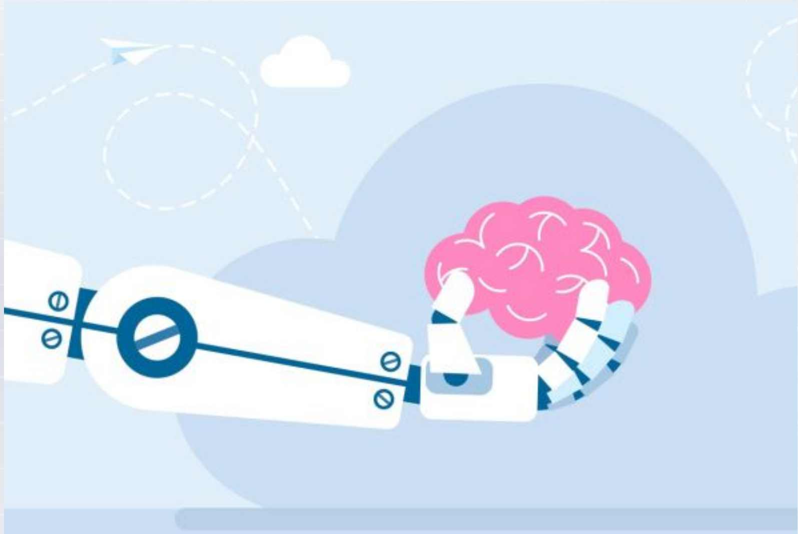
[1] Wang, Y., et al. (2023). "Improving Clinical Decision Support with Chain-of-Thought Prompting." Health Informatics Journal , 29(2), 99-114.

FUTURE APPLICATIONS

- Involve the integration of these prompting techniques into existing clinical decision support systems (CDSS)
- Use such prompting techniques for LLMs in other medical fields
    - Anything that generates summaries (patient encounters)

FUTURE RESEARCH

- Assess impact of combining RAG + Chain of Thought Prompting into a single training technique
    - Allow for more nuanced information retrieval
    - Mitigate hallucinations to percentages below 5% → usable for psychologists



Thank You for Listening!

- Thank you to Prof. Siddharth Krishnan and Billy Gao for their support
- Thank you to Indigo for the opportunity to present my research.